

Hong Kong Market Data Exploration User Manual

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1. System Overview

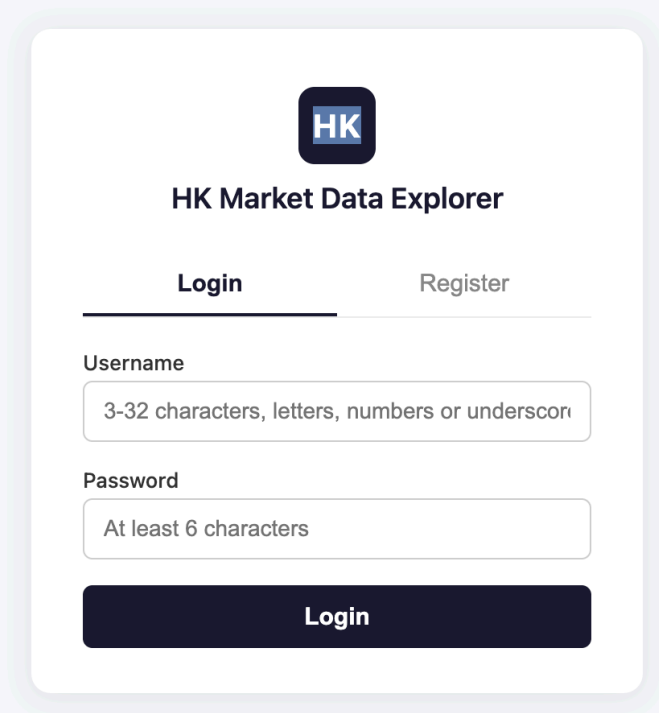
Hong Kong Market Data Exploration is a set of AI analysis tools tailored for Hong Kong market data. Users can pose questions in natural language, and the system automatically understands the questions, generates SQL, queries data, and returns text answers, charts, and table results.

- Supports data exploration of stocks, indices, industries, market capitalization, news announcements, financial statements, etc. in the Hong Kong market.
- Supports switching between Chinese and English interfaces, and query issues can be directly described using natural Chinese language.
- Supports query scope limitation by data table, suitable for demonstration, verification, and business analysis scenarios.

2. Login, Register, and Logout

When using the system for the first time, you need to register an account first. If you already have an account, you can log in directly. After logging in, the system will display the current username and you can log out via the top button.

1. Open the system address and enter the login page.
2. If you already have an account, enter your username and password, then click "Login".
3. If you need a new account, switch to "Register", enter your username, password, and confirm password, then submit.
4. After logging in, enter the main interface; after use, click "Exit" at the top.



The screenshot shows a login and registration interface for 'HK Market Data Explorer'. At the top is a logo with 'HK' in a blue square. Below the logo, the title 'HK Market Data Explorer' is centered. There are two tabs: 'Login' (active, underlined) and 'Register'. Under the 'Login' tab, there are two input fields: 'Username' with a hint '3-32 characters, letters, numbers or underscore' and 'Password' with a hint 'At least 6 characters'. Below these fields is a dark blue 'Login' button.

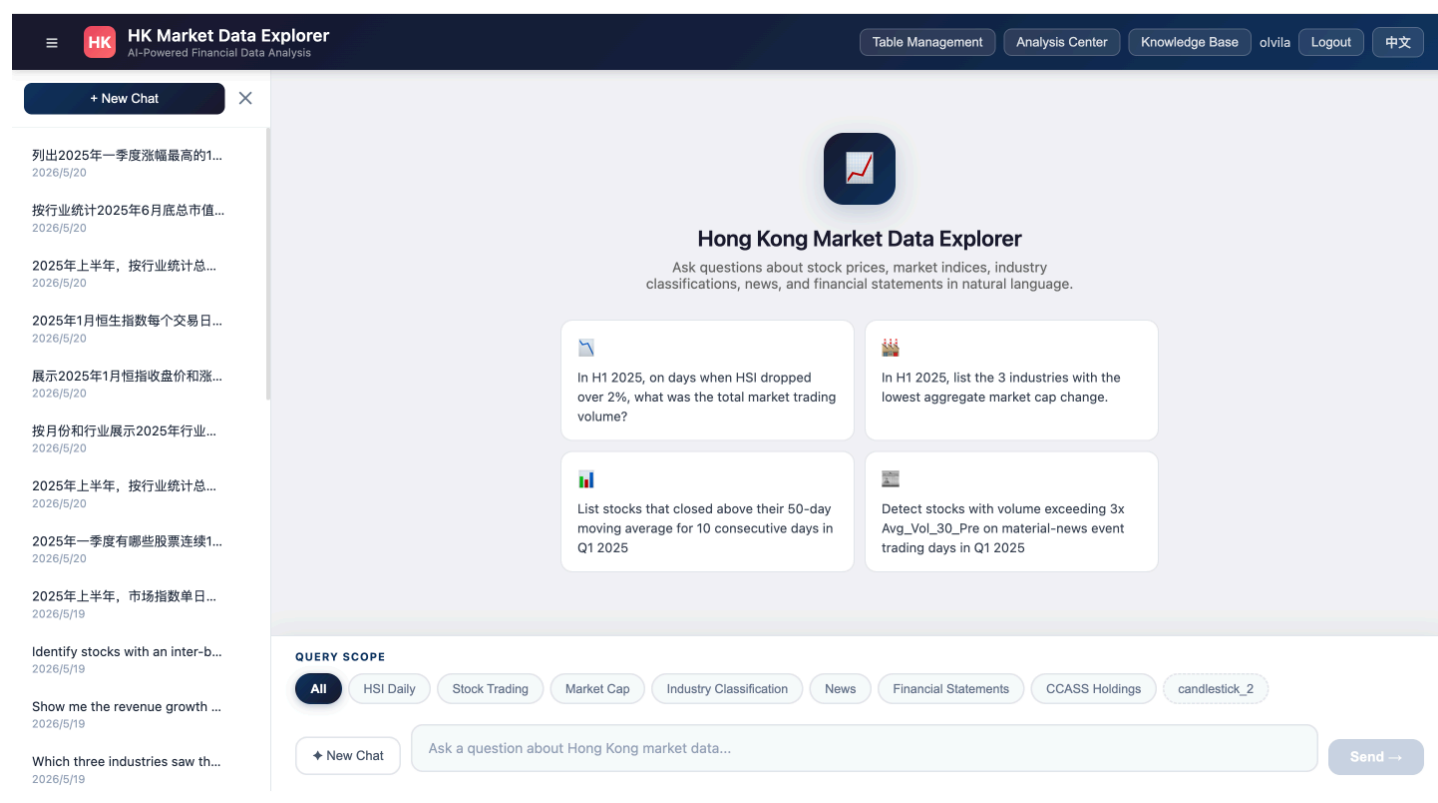
Login and Registration Page

3. Main Interface Instructions

The main interface consists of a top navigation bar, a left-side conversation list, and a middle conversation workspace. The top is used to access system-level functions; the left side is used to

manage historical conversations; and the middle area is used for asking questions and viewing results.

- Top Navigation: Displays system name, data table management, Knowledge Base, current user, and language switch.
- Left-side conversation: You can create a new conversation, select a historical conversation, and delete conversations that are no longer needed.
- Middle Workspace: Displays the welcome page, sample questions, query input box, and query results.



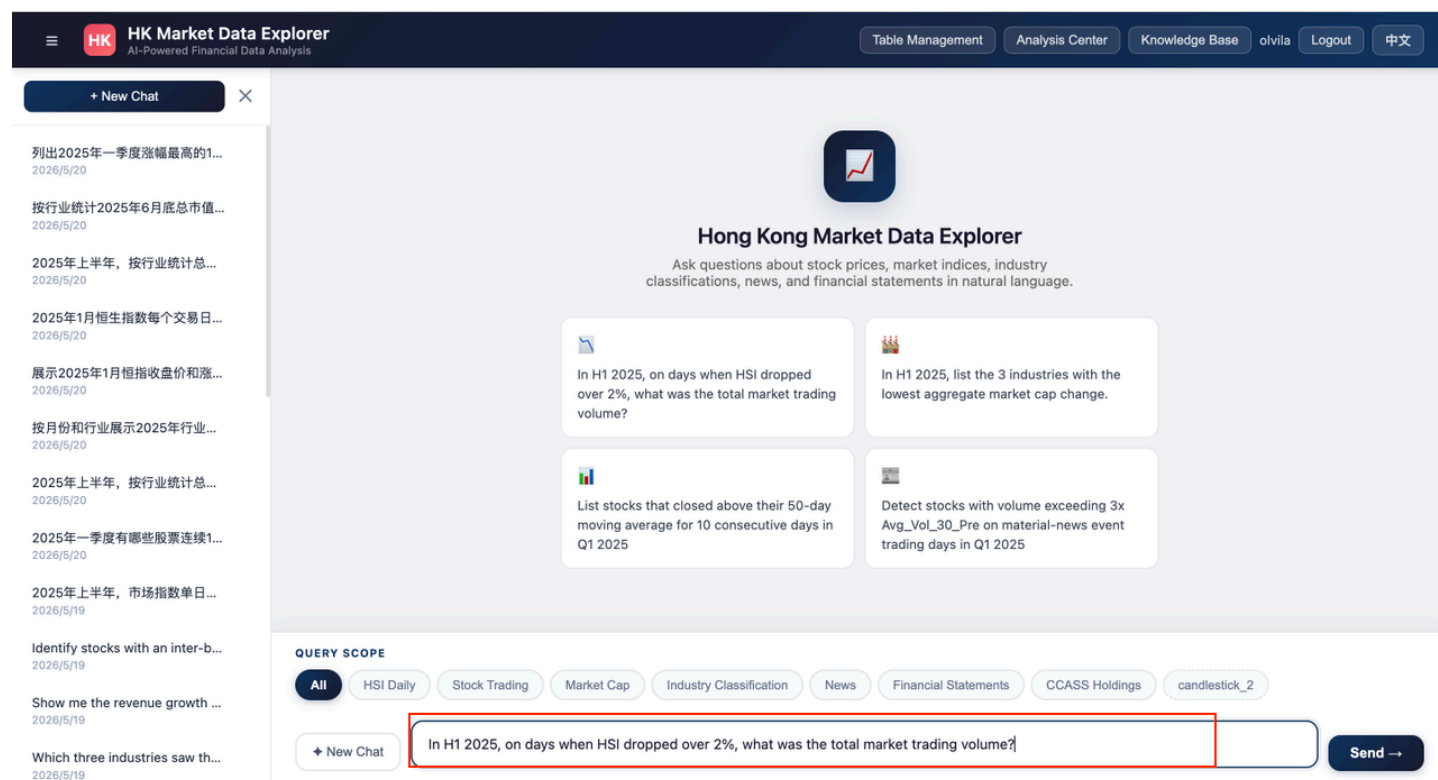
Main Interface and Welcome Page

4. Initiate a natural language query

Users can directly describe their analysis requirements in the input box, such as querying the closing price of an index, changes in industry market capitalization, stock trends, or indicators related to news announcements within a certain period of time.

1. Enter your question in the bottom input box.
2. If you need to limit the data scope, you can select one or more data tables in the "Query Scope"; if no selection is made, all available tables will be queried by default.
3. Click "Send", or press Enter in the input box to submit the question.
4. After the system starts analyzing, it will display the status of various stages such as understanding the problem, generating solutions, querying data, and generating answers.

Example Question: What was the total trading volume of the entire market on trading days when the daily decline of the market index exceeded 2% in the first half of 2025?

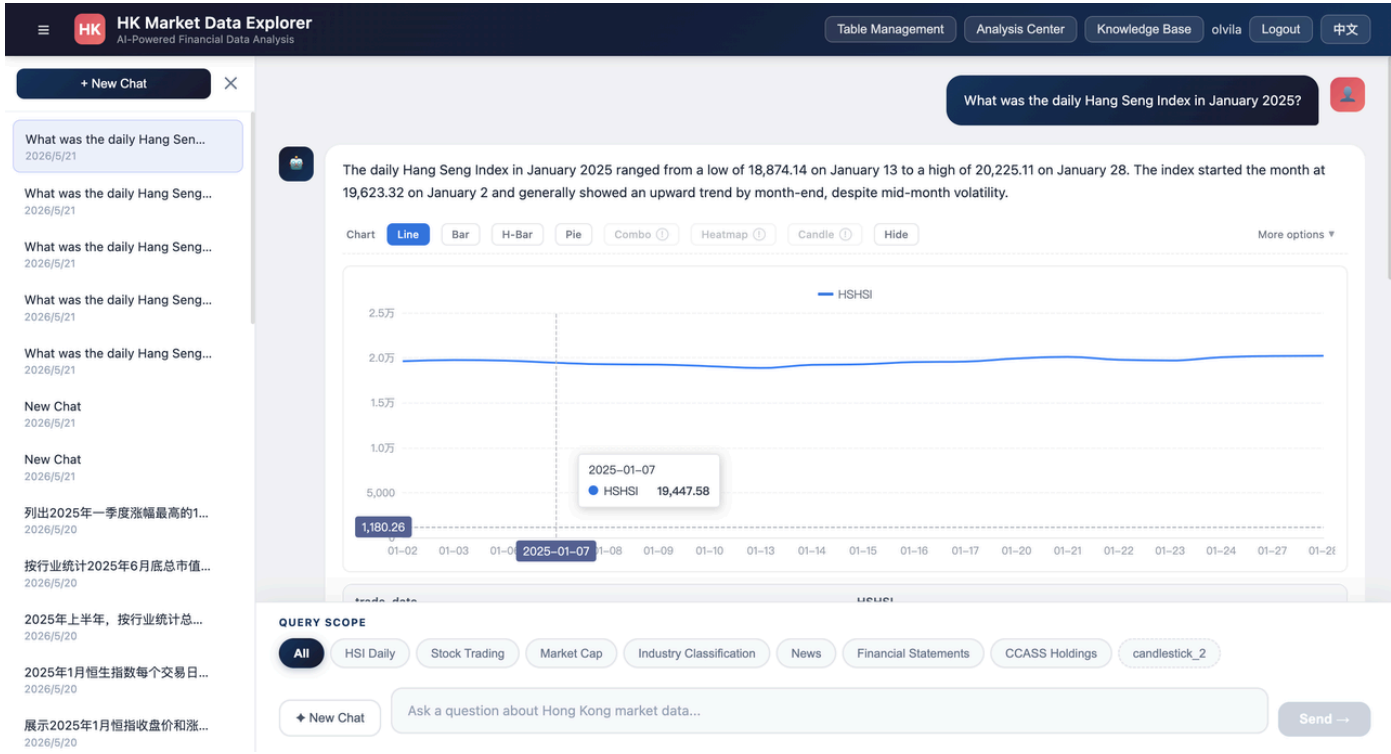


Natural Language Query

5. View the query results

After the query is completed, the system will return natural language answers in the conversation and display charts and detailed tables based on the result data. For results involving data processing, the system will provide explanations of the processing criteria.

- Text Response: Summarize the query results and key conclusions.
- Chart Area: Suitable for visual display of trend, comparison, and distribution results.
- Data Table: Displays query details and supports pagination for viewing. (Click "Hide" to switch from the chart to the data table)
- SQL Expansion: Click "Show SQL" to view the SQL actually executed by the system and copy it.
- Data processing instructions: When the results use pre-calculated or processed fields, you can expand to view the calculation criteria and core code.



Query Results - Chart

HK Market Data Explorer

Table ManagementAnalysis CenterKnowledge BaseoliviaLogout中文

+ New Chat

What was the daily Hang Sen...
2026/5/21

What was the daily Hang Seng...
2026/5/21

What was the daily Hang Seng...
2026/5/21

What was the daily Hang Seng...
2026/5/21

What was the daily Hang Seng...
2026/5/21

New Chat
2026/5/21

New Chat
2026/5/21

列出2025年一季度的涨幅最高的1...
2026/5/20

按行业统计2025年6月底总市值...
2026/5/20

2025年上半年，按行业统计总...
2026/5/20

2025年1月恒生指数每个交易日...
2026/5/20

展示2025年1月恒指收盘价和涨...
2026/5/20

按月份和行业展示2025年行业...
2026/5/20

2025年上半年，按行业统计总...
2026/5/20

2025年一季度有哪些股票连续1...
2026/5/20

What was the daily Hang Seng Index in January 2025?

The daily Hang Seng Index in January 2025 ranged from a low of 18,874.14 on January 13 to a high of 20,225.11 on January 28. The index started the month at 19,623.32 on January 2 and generally showed an upward trend by month-end, despite mid-month volatility.

Chart

LineBarH-BarPieComboHeatmapCandleHide

trade_dateHSHSI

2025-01-02	19623.3200
2025-01-03	19760.2700
2025-01-06	19688.2900
2025-01-07	19447.5800
2025-01-08	19279.8400
2025-01-09	19240.8900
2025-01-10	19064.2900
2025-01-13	18874.1400
2025-01-14	19219.7800
2025-01-15	19286.0700
2025-01-16	19522.8900
2025-01-17	19584.0600
2025-01-20	19925.8100
2025-01-21	20106.5500
2025-01-22	19778.7700
2025-01-23	19799.5200

QUERY SCOPE

AllHSI DailyStock TradingMarket CapIndustry ClassificationNewsFinancial StatementsCCASS Holdingscandlestick_2

+ New ChatAsk a question about Hong Kong market data...Send -->

Query Results – Data Table

> This query used 2 data transformation(s) ▾

HSI Daily Close

HSI Daily Trading Date

HSI Daily Close

ms_v_stk_hsi_daily.HSHSI

METHODOLOGY

This column comes from the daily closing record in the raw Hang Seng Index quote table. Import first parses the raw HSI quote date into a trading date, then keeps daily closing records to build a one-row-per-day HSI daily table for direct index queries.

CORE CODE

```
CREATE TABLE ms_v_stk_hsi_daily AS
SELECT trade_date, HSHSI, HSFIN, HSUTL, HSPROP, HSCANI,
      (HSHSI - LAG(HSHSI) OVER (ORDER BY trade_date))
      / LAG(HSHSI) OVER (ORDER BY trade_date) * 100 AS hsi_pct_change
FROM ms_t_stk_hsi
WHERE CLOSING = 9
ORDER BY trade_date
```

<> Show SQL ▾

Inaccurate result?

Data Processing Instructions

<> Hide SQL ▾

SQL

```
SELECT trade_date, HSHSI
FROM ms_v_stk_hsi_daily
WHERE trade_date >= '2025-01-01' AND trade_date <= '2025-01-31'
ORDER BY trade_date ASC
```

Copy

Inaccurate result?

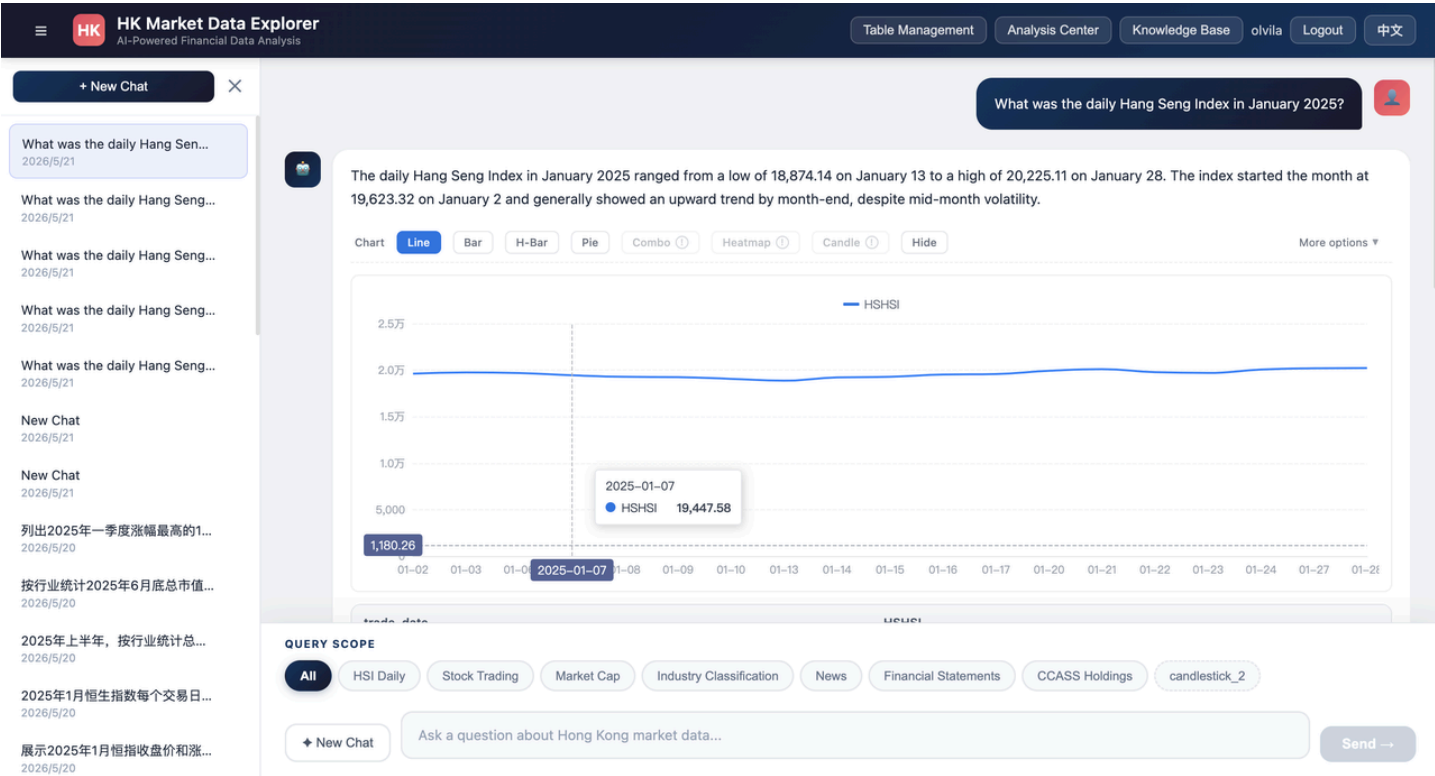
SQL Expand

6. Chart Configuration

The system supports two ways to set up charts: one is to only specify the data to be queried when asking a question, and the system will automatically recommend a chart after querying the data; the other is [to specify the chart to be generated when querying](#) or [to continue asking follow-up questions based on existing query results](#), such as "change to a bar chart", "only show the top 10", "show data labels", and the system will preferentially reuse the previous round's results to redraw the chart.

- Chart Type: Supports line, column, bar, pie, Combo, heatmap, and candlestick charts. You can also click "Hide Chart" to switch to a data table.
- Dimensions and Metrics: Select the horizontal axis dimension, numerical metric, and optional legend field.
- Combo Chart: Can switch between bar or line for different indicators and set the primary/secondary Y-axis.
- Sorting and Top N: Can sort in ascending or descending order by a specified field and limit the display to the top N entries.
- Display Options: Line and Combo charts can display marker points, and all charts can display data labels as needed.

6.1 Automatic Chart Recommendation



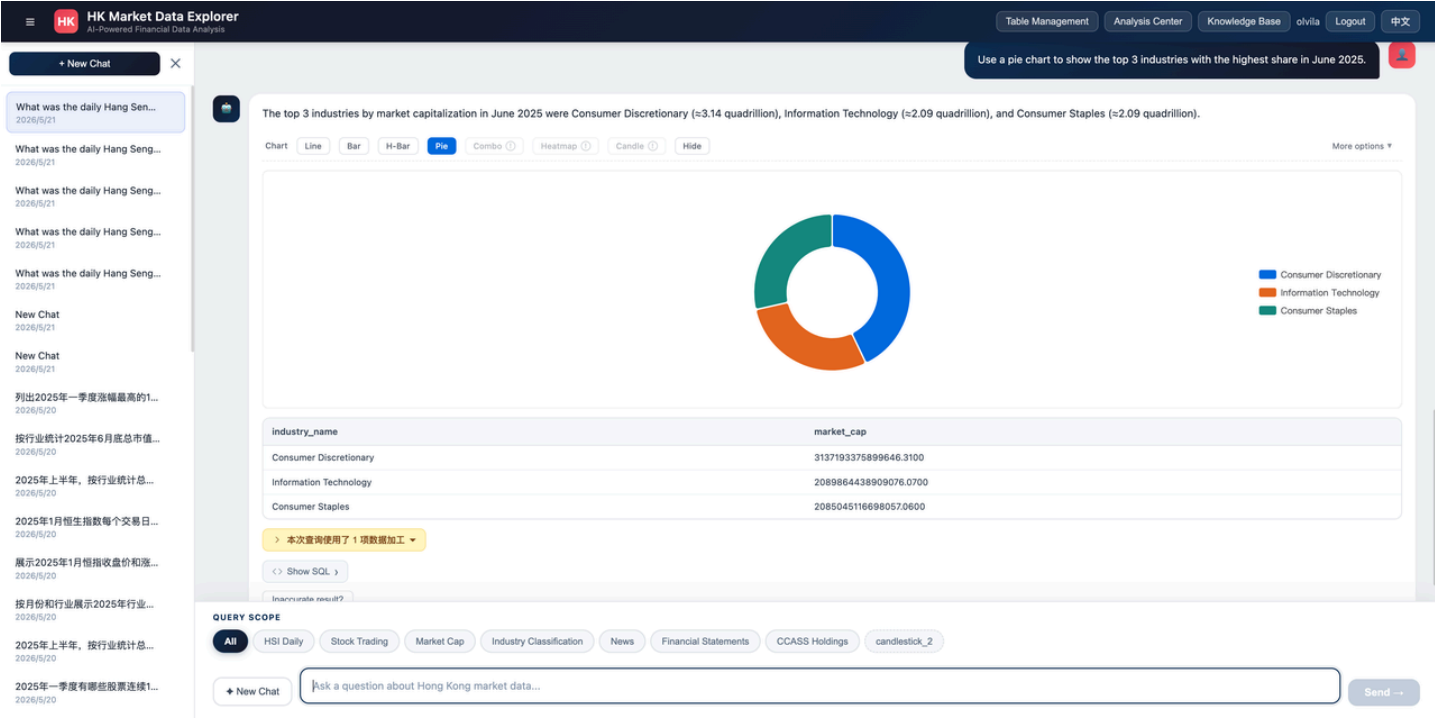
Automatically Recommended Chart

6.1 Request image generation using natural language

When asking questions, it is recommended to specify the analysis object, time range, indicator, grouping dimension, and chart type simultaneously. The more complete the information, the easier it is for the system to return results that can be directly plotted.

Chart Type	Applicable Scenarios	Recommended Questioning Method
Line	Displays time series trends, such as the change of indicators over dates.	What was the Closing Price of CKH HOLDINGS on each trading day in January 2025? Please use a line chart to show the trend.
Bar	Compare the numerical values of the indicator under different dimensional values.	In the first half of 2025, calculate the change in total market capitalization by industry and present it using a bar chart.
H-Bar	Category names that are long or ranking results are more clearly presented horizontally.	List the top 10 stocks with the highest increase in the first quarter of 2025 and display them using a bar chart.

Pie	Show the composition proportion (the number of categories should not be too large).	Use a pie chart to show the top 3 industries with the highest proportion in June 2025.
Combo	Displays more than two indicators simultaneously, each with its own measurement axis.	Display the daily Hang Seng Index and daily percentage change for each day in March 2025 using a Combo chart, with the index represented by columns and the daily percentage change by a line.
Heatmap	Cross-analysis of an indicator across two dimensions.	The monthly maximum closing prices of CKH HOLDINGS in 2025 and 2026 are presented using a heatmap.
Candlestick	Displays four price fields: Opening Price, Closing Price, High Price, and Low Price.	What were the opening price, closing price, highest price, and lowest price of XXXX stock on XX/XX/XXXX? Show it using a candlestick chart.

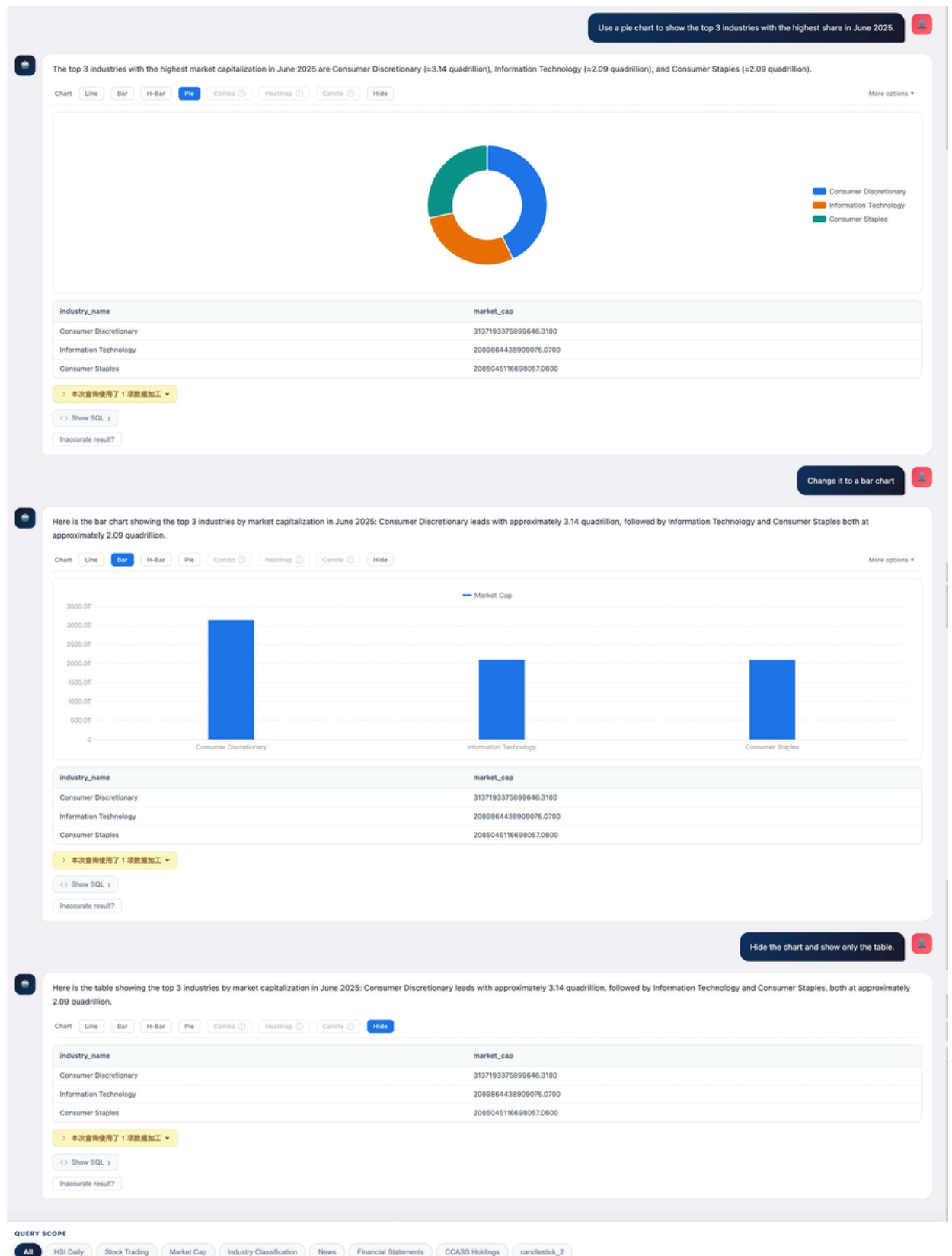


Specify Chart Type

6.2 Naturally inquire about image modification based on existing results using natural language

Within the same conversation, the system will retain the most recent response with query results. When the user subsequently enters short phrases such as "change to a bar chart", "switch to a pie chart", "only show the top 10", "add data labels", etc., the system will recognize them as follow-up questions to the previous round of results and prioritize reusing the existing results to regenerate the chart.

1. First initiate a complete query and have the system return data tables and charts.
2. Continue to input requirements for modifying the chart in the same conversation, for example, "change to a bar chart".
3. The system returns a new response and chart configuration, usually without re-executing the SQL query.
4. If the natural language image modification does not fully meet expectations, you can manually adjust it using the chart toolbar.



Natural language follow-up, change to a bar chart, then switch to a data table

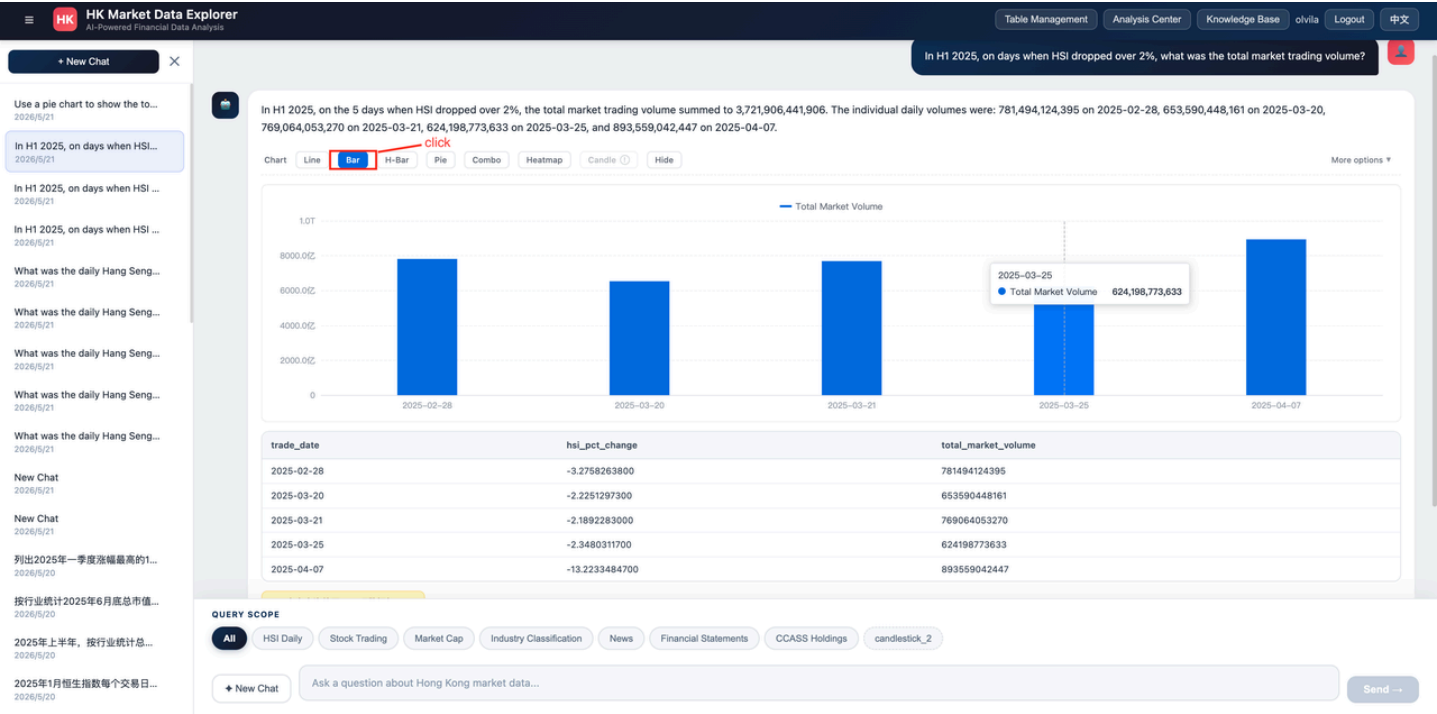
Examples of common follow-up questions:

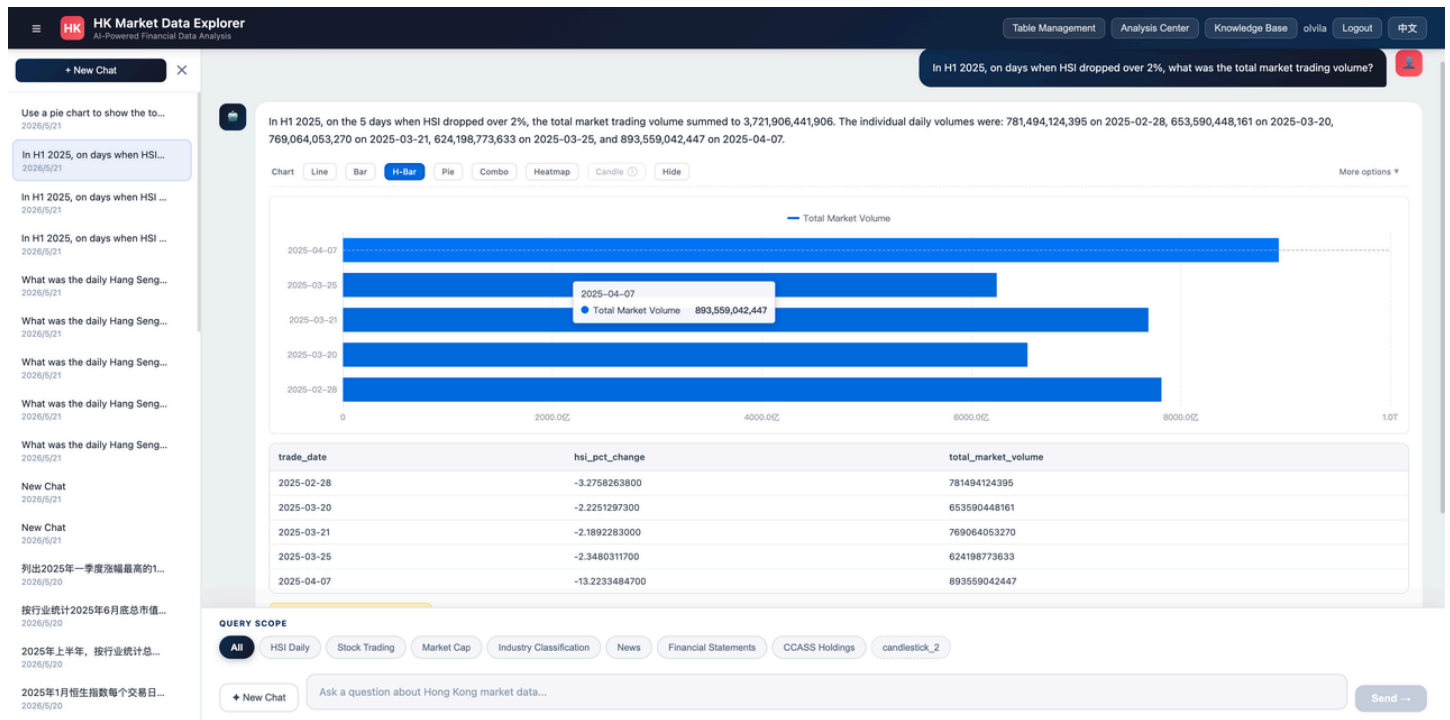
- "Change to bar chart"
- "Change to bar chart, sorted in descending order by value"

- "Only look at Top 10"
- "Use a pie chart to show the proportion"
- "Place the trading volume on the right axis and display it using a Combo chart"
- "Hide the chart and only view the table"

6.3 Manually Switch Chart

Clicking the chart type button can switch between line, column, bar, pie, Combo, heatmap, and candlestick charts, and clicking "Hide" switches to the data table. If a chart type button is grayed out, it indicates that the number of columns in the current query result is insufficient. For example, a Combo chart requires at least one dimension and two metrics, a heatmap requires at least two dimensions and one metric, and a candlestick chart requires five types of fields: time, open, close, high, and low.

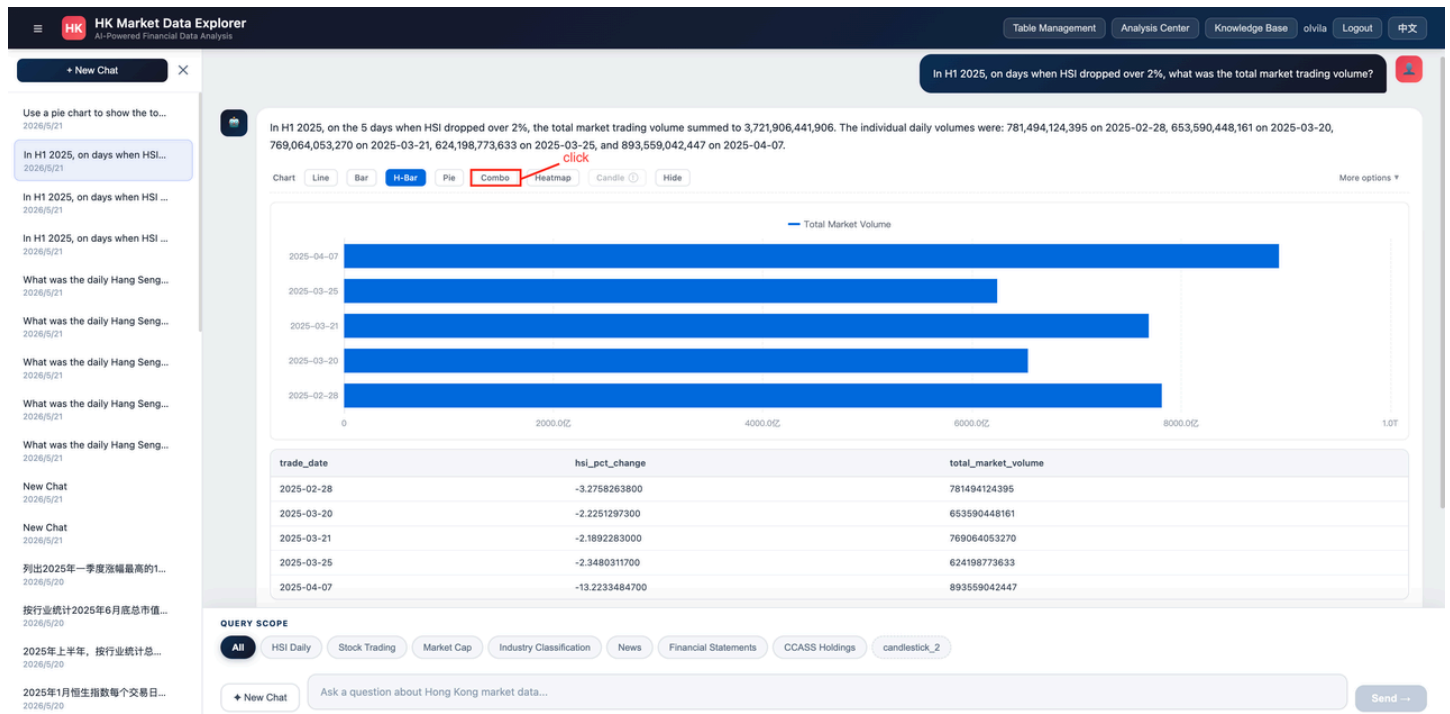


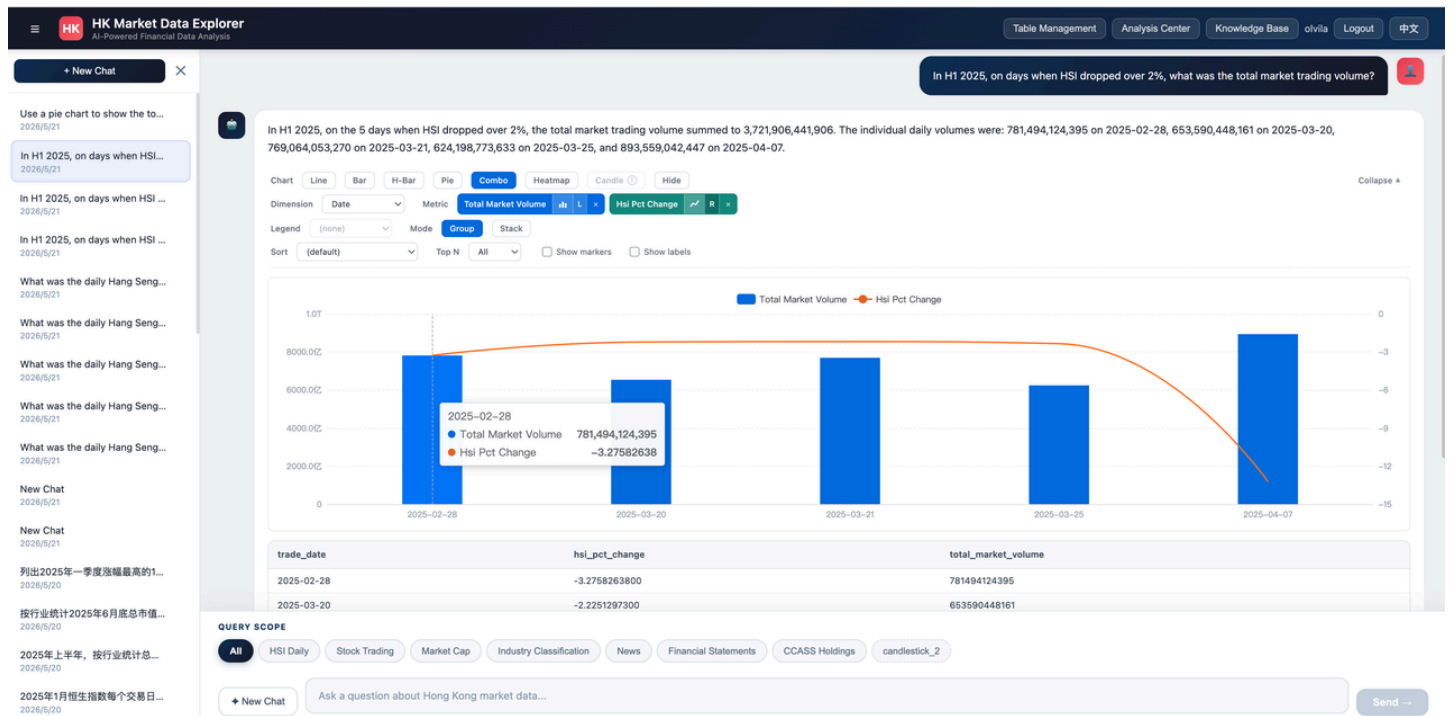
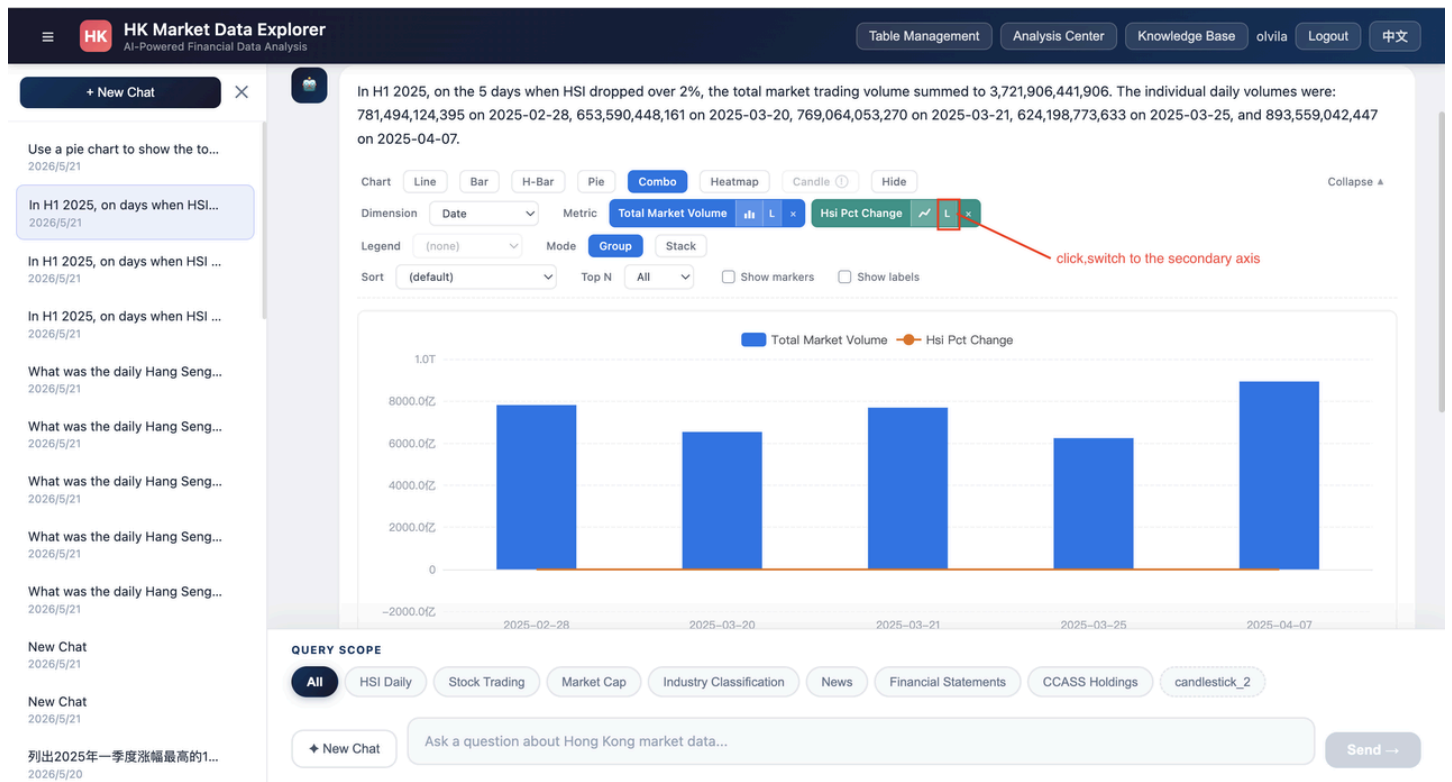


Manually switch the bar chart to a column chart

6.4 Manually Configure Chart

Based on the current chart, after clicking "More Configurations", you can manually configure fields and adjust display rules.





Configuration Combo Diagram

- Dimension: Usually serves as the horizontal axis or categorical field, such as date, industry, stock code.
- Indicators: Used to plot numerical values, such as Closing Price, Trading Volume, Market Capitalization, and Growth Rate.
- Legend: Used to split the same indicator into multiple lines or groups of columns based on another field.
- Sorting: After selecting the sorting field, you can switch between ascending or descending order.

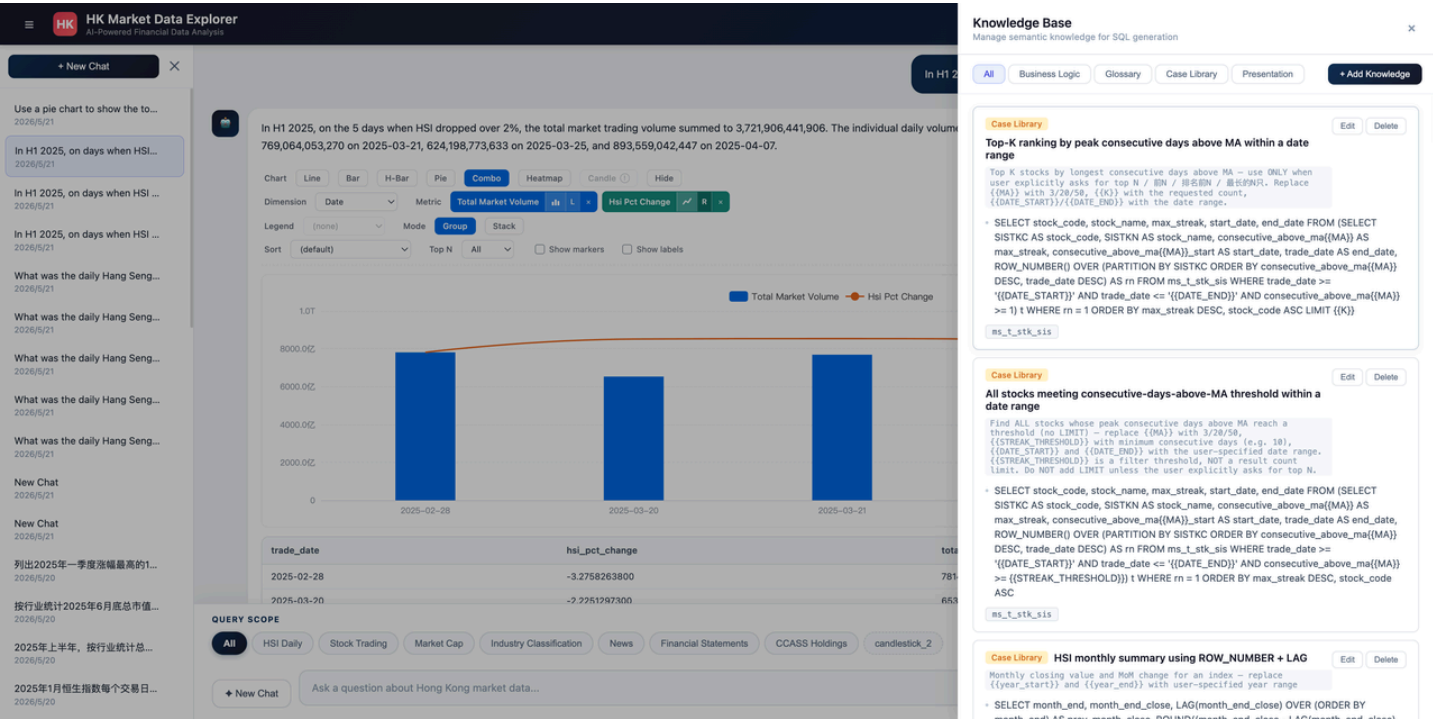
- Top N: Limit display to all, top 5, top 10, top 20, or top 50 items.
- Show Markers: Applies to the lines in line charts or Combo charts.
- Show Data Labels: Display values directly on the chart.

7. Knowledge Base

The Knowledge Base is used to maintain semantic knowledge related to SQL generation, including business logic, glossaries, case libraries, and display rules. The maintained knowledge will help the system understand data models and business specifications more accurately.

[Do not edit or modify the existing Knowledge Base configuration in the current environment]

1. Click on "Knowledge Base" at the top to enter the Knowledge Base panel.
2. Filter by the top labels for All, Business Logic, Glossary, Case Library, or Display Rules.
3. Click "Add Knowledge" and fill in the type, identifier, name, content, and associated table.
4. Existing entries can be modified by clicking "Edit" or removed by clicking "Delete".



Knowledge Base Management

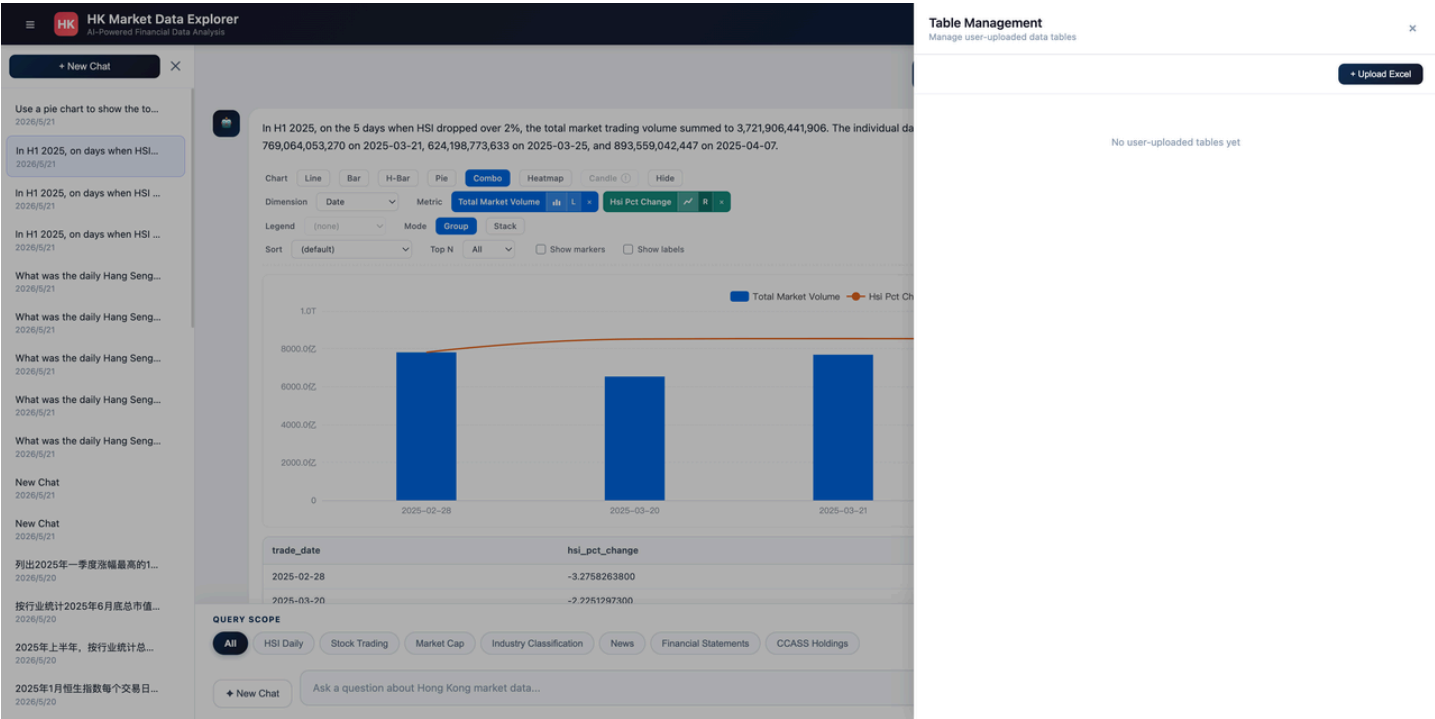
8. Table Management

8.1Table Management

Data table management is used to maintain data tables uploaded by users. The system supports uploading Excel or CSV files, creates tables after parsing fields, and allows editing the metadata

of tables and columns.

1. Click "Data Table Management" at the top to open the sidebar panel.
2. View existing user tables, including table name, table comment, number of rows, and creation time.
3. Click "Preview" to view sample data in the table.
4. Click "Edit Metadata" to maintain table and column comments.
5. Click "Delete" to delete the user table, and the system will prompt for confirmation before deletion.



Data Table Management

8.2 Upload Excel and Analyze Data

1. Click "Upload Excel" in the data table management panel.
2. Drag a.xlsx,.xls, or.csv file into the upload area, or click the area to select a local file.
3. After the system uploads and parses the file, it will display the table name, table comment, field type, and preview data.
4. Confirm or modify the table name, field type, and column comment.
5. Click "Confirm Table Creation" and wait for the import to complete.

Table names only allow lowercase letters, numbers, and underscores. It is recommended to fill in clear comments for tables and fields so that they can be more easily understood by the system during subsequent natural language queries.

HK

HK Market Data Explorer

AI-Powered Financial Data Analysis

+ New Chat

X

Use a pie chart to show the to...

2026/5/21

In H1 2025, on days when HSI...

2026/5/21

In H1 2025, on days when HSI ...

2026/5/21

In H1 2025, on days when HSI ...

2026/5/21

What was the daily Hang Seng...

2026/5/21

What was the daily Hang Seng...

2026/5/21

What was the daily Hang Seng...

2026/5/21

What was the daily Hang Seng...

2026/5/21

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2026/5/21

New Chat

2026/5/21

New Chat

2026/5/21

列出2025年一季度涨幅最高的1...

2026/5/20

按行业统计2025年6月底总市值...

2026/5/20

2025年上半年，按行业统计总...

2026/5/20

2025年1月恒生指数每个交易日...

2026/5/20

In H1 2025, on the 5 days when HSI dropped over 2%, the total market trading volume summed to 3,721,906,441,906. The individual da

769,064,053,270 on 2025-03-21, 624,198,773,633 on 2025-03-25, and 893,559,042,447 on 2025-04-07.

Chart

Line

Bar

H-Bar

Pie

Combo

Heatmap

Candle

Hide

Dimension

Date

Metric

Total Market Volume

↓

↑

HSI Pct Change

✓

↓

↑

R

+

Legend

[none]

Mode

Group

Stack

Sort

(default)

Top N

All

☐ Show markers

☐ Show labels

1.0T

8000.0CZ

8000.0CZ

4000.0CZ

2000.0CZ

0

2025-02-28

2025-03-20

2025-03-21

trade_date

hsi_pct_change

2025-02-28

-3.2756263800

2025-03-20

-2.2251297300

QUERY SCOPE

All

HSI Daily

Stock Trading

Market Cap

Industry Classification

News

Financial Statements

CCASS Holdings

+ New Chat

Ask a question about Hong Kong market data...

Table Management

Manage user-uploaded data tables

X

+ Upload Excel

No user-uploaded tables yet

HK

HK Market Data Explorer

AI-Powered Financial Data Analysis

+ New Chat

X

Use a pie chart to show the to...

2026/5/21

In H1 2025, on days when HSI ...

2026/5/21

In H1 2025, on days when HSI ...

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What was the daily Hang Seng...

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What was the daily Hang Seng...

2026/5/21

New Chat

2026/5/21

New Chat

2026/5/21

列出2025年一季度涨幅最高的1...

2026/5/20

按行业统计2025年6月底总市值...

2026/5/20

2025年上半年，按行业统计总...

2026/5/20

2025年1月恒生指数每个交易日...

2026/5/20

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Chart

Line

Bar

H-Bar

Pie

Combo

Heatmap

Candle

Hide

Dimension

Date

Metric

Total Market Volume

↓

↑

HSI Pct Change

✓

↓

↑

R

+

Legend

[none]

Mode

Group

Stack

Sort

(default)

Top N

All

☐ Show markers

☐ Show labels

1.0T

8000.0CZ

8000.0CZ

4000.0CZ

2000.0CZ

0

2025-02-28

2025-03-20

2025-03-21

trade_date

hsi_pct_change

2025-02-28

-3.2756263800

2025-03-20

-2.2251297300

QUERY SCOPE

All

HSI Daily

Stock Trading

Market Cap

Industry Classification

News

Financial Statements

CCASS Holdings

+ New Chat

Ask a question about Hong Kong market data...

Table Management

Manage user-uploaded data tables

X

+ Upload Excel

No user-uploaded tables yet

Upload Excel

Table Name

candlestick

Only lowercase letters, numbers, and underscores

Table Description

Column

Column

Type

Comment

日期

DATE

股票代码

BIGINT

公司名称

VARCHAR(255)

开盘价

DECIMAL(18,6)

收盘价

DECIMAL(18,6)

最高价

DECIMAL(18,6)

最低价

DECIMAL(18,6)

Preview Data (31 Rows)

日期	股票代码	公司名称	开盘价	收盘价	最高价	最低价
2025-01-01	001	新华电源	30.50	30.80	31.00	30.40
2025-01-02	001	新华电源	30.80	31.20	31.50	30.75
2025-01-03	001	新华电源	31.20	31.10	31.40	30.90
2025-01-04	001	新华电源	31.10	30.95	31.25	30.80
2025-01-05	001	新华电源	30.95	31.30	31.45	30.90
2025-01-06	001	新华电源	31.30	31.60	31.80	31.20
2025-01-07	001	新华电源	31.60	31.40	31.70	31.30

Cancel

Create Table

HK Market Data Explorer
AI-Powered Financial Data Analysis

Table Management
Manage user-uploaded data tables

candlestick
31 Rows - 2025-05-21 06:18:58

Hong Kong Market Data Explorer
Ask questions about stock prices, market indices, industry classifications, news, and financial statements in natural language.

QUERY SCOPE
All HSI Daily Stock Trading Market Cap Industry Classification News Financial Statements CCASS Holdings candlestick

New Chat
Ask a question about Hong Kong market data...

HK Market Data Explorer
AI-Powered Financial Data Analysis

Table Management **Analysis Center** **Knowledge Base** **oivia** **Logout** **中文**

New Chat
Ask a question about Hong Kong market data...

Hong Kong Market Data Explorer
Ask questions about stock prices, market indices, industry classifications, news, and financial statements in natural language.

QUERY SCOPE
All HSI Daily Stock Trading Market Cap Industry Classification News Financial Statements CCASS Holdings candlestick

New Chat
Ask a question about Hong Kong market data...

What were the closing prices of the Hang Seng Index for each trading day in January 2025? Please show the trend using a line chart.

Send

Upload the table file and analyze the data

9. Notes

- The problem description should include time range, indicator name, sorting method, and quantity limit as much as possible.
- If the problem only involves certain tables, it is recommended to first select the relevant tables in the query scope to reduce system understanding bias.
- When the query result is empty, you can check whether the time range, company name, stock code, or industry name is consistent with the data coverage scope.

- When the chart fields do not match, you can reselect dimensions and metrics in the chart toolbar.
- After uploading the data table, it is recommended to promptly complete the table and column comments to improve the effectiveness of subsequent queries.